Introduction

Banking institutions are avid users of data to inform the evolution of their products and tailoring of financial solutions to their customers, to aid attraction and retention (PricewaterhouseCoopers, no date). A Portuguese bank conducted a telemarketing campaign to convince its clients to open a long-term deposit account. The dataset (mma), collated by Moro, Cortez and Rita (2014) included 4100 observations collected between 2008 and 2013, and contains twenty variables listed in appendix A.

The aim of this programming exercise is to construct a predictive model capable of identifying which clients will agree to open an account, thus improving the efficiency of subsequent marketing campaigns. This report includes an exploratory data analysis (EDA) of the dataset to identify which variables should and should not be included in the model, the development of two potential models and an evaluation of each to conclude which model is able to reliably predict the outcome.

Python was used to conduct the EDA and build the predictive models due to strong familiarity with the language as well as its capabilities in data cleaning using the pandas library, visualisation using matplotlib and seaborn, and statistical model development using scikit-learn.

Exploratory Data Analysis

An initial exploration of the dataset was conducted; the .info() pandas method was used to retrieve basic information including column names and data types, while using .describe() obtained summary statistics for the numerical variables such as the mean, standard deviation, and minimum and maximum figures (figure 1). A mean

	age	duration	campaign	pdays	previous \
count	4100.000000	4100.000000	4100.000000	4100.000000	4100.000000
mean	40.117073	256.754634	2.538537	960.243415	0.190732
std	10.315182	254.399637	2.568833	192.349065	0.542609
min	18.000000	0.000000	1.000000	0.000000	0.000000
25%	32.000000	103.000000	1.000000	999.000000	0.000000
50%	38.000000	181.000000	2.000000	999.000000	0.000000
75%	47.000000	317.000000	3.000000	999.000000	0.000000
max	88.000000	3643.000000	35.000000	999.000000	6.000000

	emp.var.rate	cons.price.idx	cons.conf.idx	euribor3m	nr.employed
count	4100.000000	4100.000000	4100.000000	4100.000000	4100.000000
mean	0.085171	93.580123	-40.498463	3.621421	5166.472780
std	1.563183	0.579578	4.593385	1.733708	73.664618
min	-3.400000	92.201000	-50.800000	0.635000	4963.600000
25%	-1.800000	93.075000	-42.700000	1.334000	5099.100000
50%	1.100000	93.749000	-41.800000	4.857000	5191.000000
75%	1.400000	93.994000	-36.400000	4.961000	5228.100000
max	1.400000	94.767000	-26.900000	5.045000	5228.100000

Figure 1 - Summary Statistics for Numerical Variables

client age of 40.12 indicates a client cohort that is economically active and in employment.

As Python does not contain an in-built method for categorical variables, a custom function, quick_summary(), was created to return summaries for those variables. This is particularly useful to determine the class imbalance within the target variable, ‘y’ (renamed to ‘outcome’ after initial exploration). 3649 clients responded with ‘no’ and 451 responded with ‘yes’, showing a significant imbalance (figure 2).

--- Summary for 'y' ---

y	Count	Percentage
no	3649	89.0
yes	451	11.0

Total Rows: 4100

Figure 2 - Summary Statistics for 'y' (outcome)

The quantity of unknown values for each variable was obtained (figure 3) - ‘default’ possessed the highest number of unknowns with 799; ‘marital’ contained 11.

```
marital     11
job         39
housing    104
loan       104
k          167
default    799
dtype: int64
```

Figure 3 - Unknown Values for Applicable Variables

Univariate histograms displaying the distribution of all numerical variables, shown in figure 4. Notable callouts include ‘age’ which displays skewness to the right and ‘pdays’ which possesses left skew, since most values were 999, the number used to indicate that the customer was not contacted previously. ‘Campaign’ is also skewed to the right; however, this signals that clients are not contacted on numerous occasions. Individual plots for ‘age’ and ‘pdays’ were created to obtain the true distribution of the variables excluding outliers.

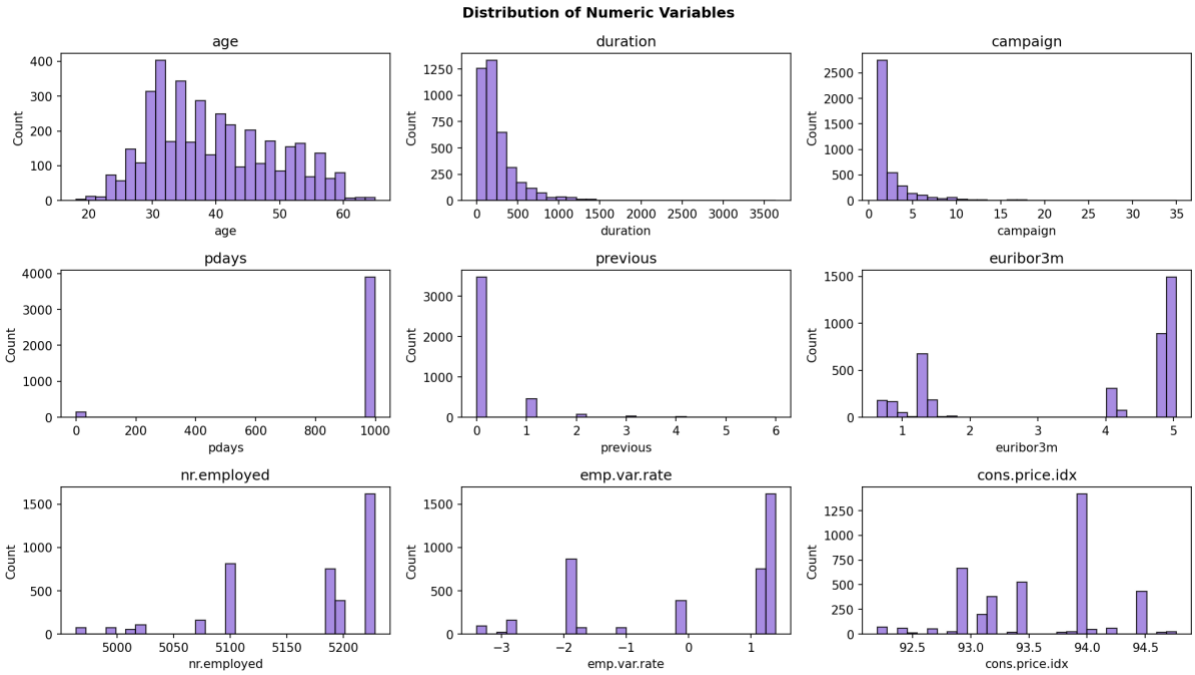

Figure 4 - Histograms Displaying Distributions of All Numerical Variables

Individual plots for ‘age’, excluding outliers over 65, and ‘pdays’ excluding 999 shows the value of visualising data before cleaning and removing outlier values. Anscombe’s Quartet reinforces this as multiple datasets with similar descriptive statistics can have vastly different plots. It also demonstrates the value of omitting outliers, crucial for developing accurate predictive models (Gough and Zhao, 2023). Figure 5 presents a less skewed distribution. Figure 6 shows a right skew with a lower number of days between contacting clients again.

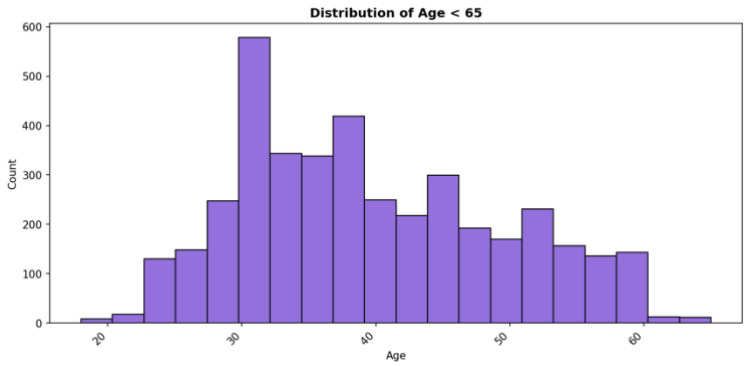

Figure 5 - Histogram Showing Distribution of 'age' Excluding Outliers

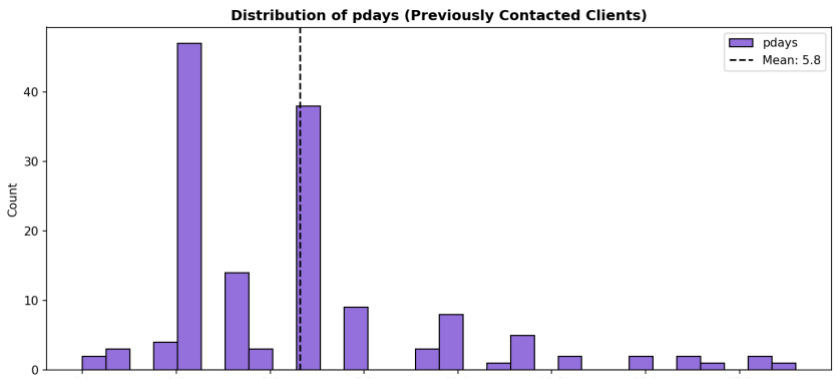

Figure 6 - Histogram Showing Distribution of 'pdays' Excluding 999

Figure 7 displays frequency bar charts for all categorical variables in a similar format to figure 3. May was the month with the highest number of contacts made and contacts were generally higher between May and August. 64.10% of contacts were made to clients’ mobile phones rather than a household or work telephone – most of

the clients were employed and would be more likely to miss telephone calls. The plot for the ‘day_of_week’ variable shows little variation between days and is unlikely to have a predictive impact. ‘Default’ is unlikely to have an impact as the vast majority of customers have not defaulted before.

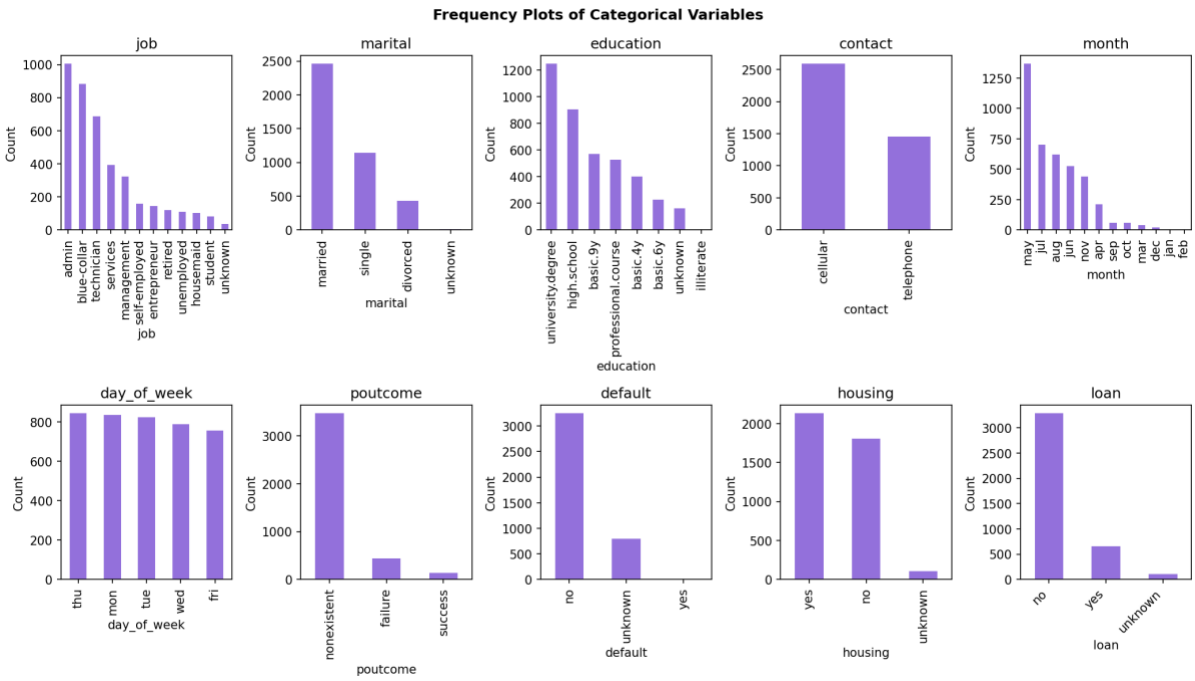

Figure 7 - Bar Plots Displaying Frequencies of All Categorical Variables

Bivariate analyses were subsequently conducted to determine the rate of clients agreeing to open an account by variable. A custom function, plot_conversion_rate(), was developed and used to create multiple bivariate plots with fewer lines of code for ‘month’, ‘poutcome’ and ‘contact’ variables.

March, December, and September had the highest conversion rates (figure 8). Figure 9 shows that clients contacted by mobile agreed to an account more than those contacted with telephones, supporting the univariate ‘contact’ plot and its contextual explanation. Figure 10 demonstrates that customers who agreed to open an account previously are willing to open another.

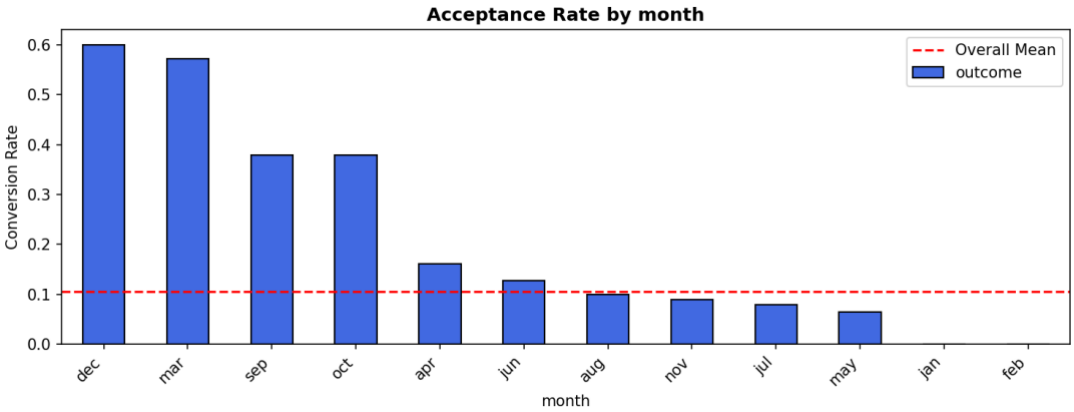

Figure 8 - Conversion Rate by Month

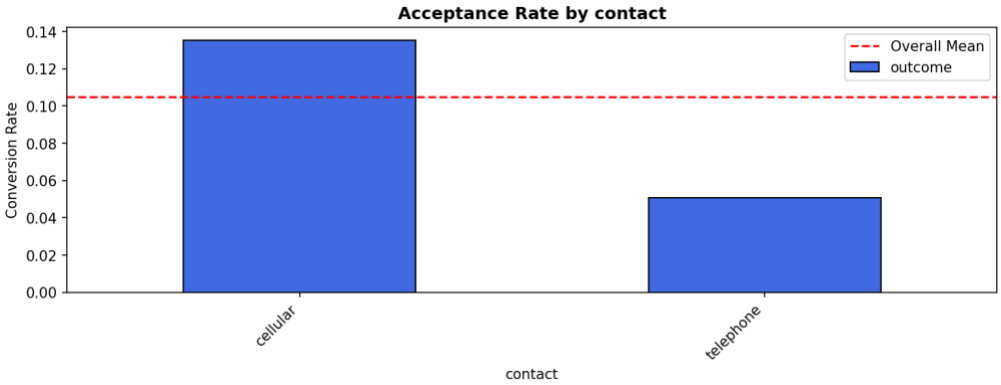

Figure 9 - Conversion Rate by Contact Method

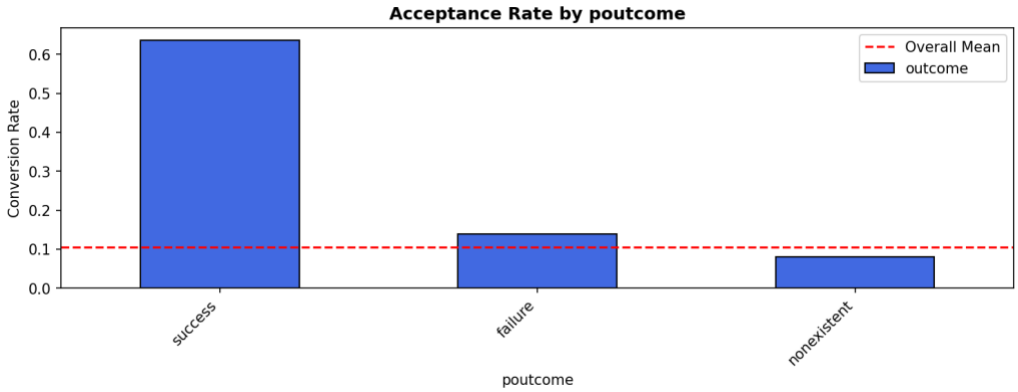

Figure 10 - Conversion Rate by Outcome of Previous Campaign

Macroeconomic indicators seem to be strong predictors of opening a long-term account, evidenced by the box plots in figure 11. More clients agreed to open an account during periods of lower interest rates, a finding supported by Staal's (2023) analysis of the effects of decreasing interest rates on household savings.

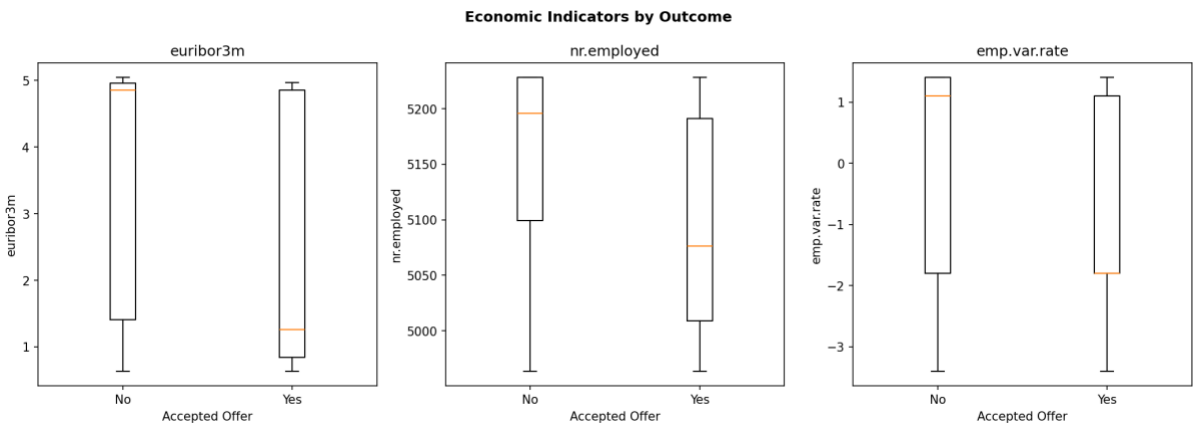

Figure 11 - Box Plots of Economic Indicators by Outcome

A correlation matrix was created, shown in figure 12, and displays strong positive correlations between the 'euribor3m', 'nr.employed' and 'emp.var.rate' variables. A consideration must be made when developing the model due to the risk of multicollinearity.

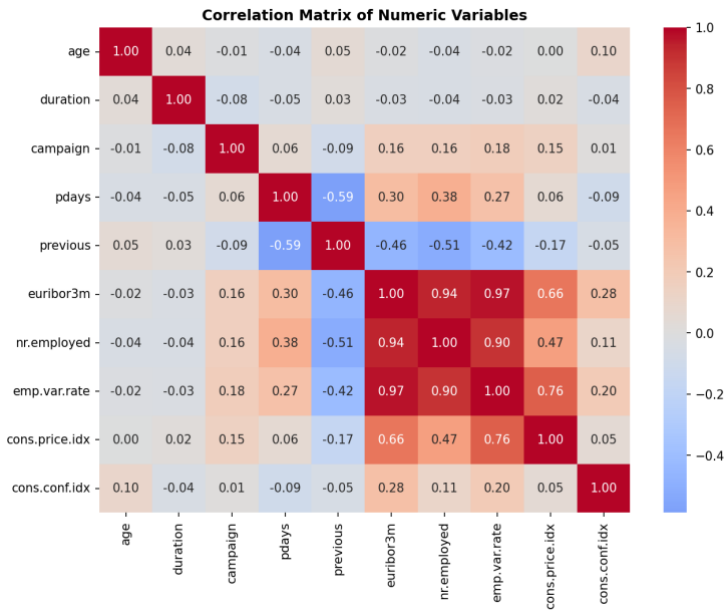

Figure 12 - Correlation Matrix of Numerical Variables

Model Development

As the dataset included multiple independent quantitative and qualitative variables, a multiple regression analysis was chosen. Logistic regression requires a binary target variable which this dataset has, making it the predictive modelling technique of choice (*What is Predictive Modeling? Types & Techniques*, no date).

'Age' values over 65 were omitted as identified in figure 5. All categorical variables were encoded as models require numeric values instead of strings. The risk of multicollinearity was avoided using the 'drop_first = True' parameter to drop all dummy columns which would introduce dummy information. An initial stratified train/test split was created to confirm the class imbalance identified earlier. Once confirmed, two approaches were tested to determine which method will be used to balance classes: class_weight and Synthetic Minority Over-sampling Technique (SMOTE). The class_weight parameter alters the model's learning while SMOTE modifies the training data.

Recall scores were better using the class_weight approach (0.86) versus SMOTE (0.57) and will be used to balance the classes in the final models.

The first model, 'model1', was constructed and included all variables. The F1-score for the overall model is 0.61, with a recall score of 0.86. Given the class imbalance, it is misleading to rely solely on the recall score. When split out by outcome, 'yes' has an F1-score of 0.61 versus 0.93 for 'no'. This can be interpreted to mean that the model correctly predicts that a customer will agree to open an account 61% of the

time. This model includes all variables, including ‘duration’, which is only known after the call with the customer takes place. This undermines the reliability of the model and is likely to be one of the reasons behind the lower F1-score for ‘yes’ (Bouke, Zaid and Abdullah, 2024). This model’s accuracy may also be impacted by multicollinearity occurring between variables with a strong correlation, such as ‘euribor3m’, ‘nr.employed’ and ‘emp.var.rate’.

A second model, ‘model2’, excluded ‘duration’, ‘nr.employed’ and ‘emp.var.rate’ variables, with the aim of improving the reliability and accuracy of the model. ‘Poutcome’, ‘default’ and ‘day_of_week’ were also omitted as highlighted during the univariate EDA.

The F1-score for ‘model2’ is 0.43, with a recall score of 0.60. When split by outcome, ‘yes’ had an F1-score of 0.43 versus 0.90 for ‘no’, which is a decrease compared to ‘model1’. These scores support the idea that multicollinearity between ‘euribor3m’, ‘nr.employed’ and ‘emp.var.rate’, as well as ‘duration’, ‘poutcome’, and ‘default’, inflated the previous model’s accuracy and reliability.

Model Evaluation

Further evaluative activities were conducted, including confusion matrices for both models (figure 13). False positives located on the top right of both plots indicate that time and resources would be wasted contacting clients who would not open an account despite the model predicting that they would. False negatives located on the bottom left, where the model incorrectly predicted that the client would not open an

account, would result in genuine opportunities missed. Slightly higher false positives and negatives for ‘model2’ reinforce confirm the difference in F1 and recall scores.

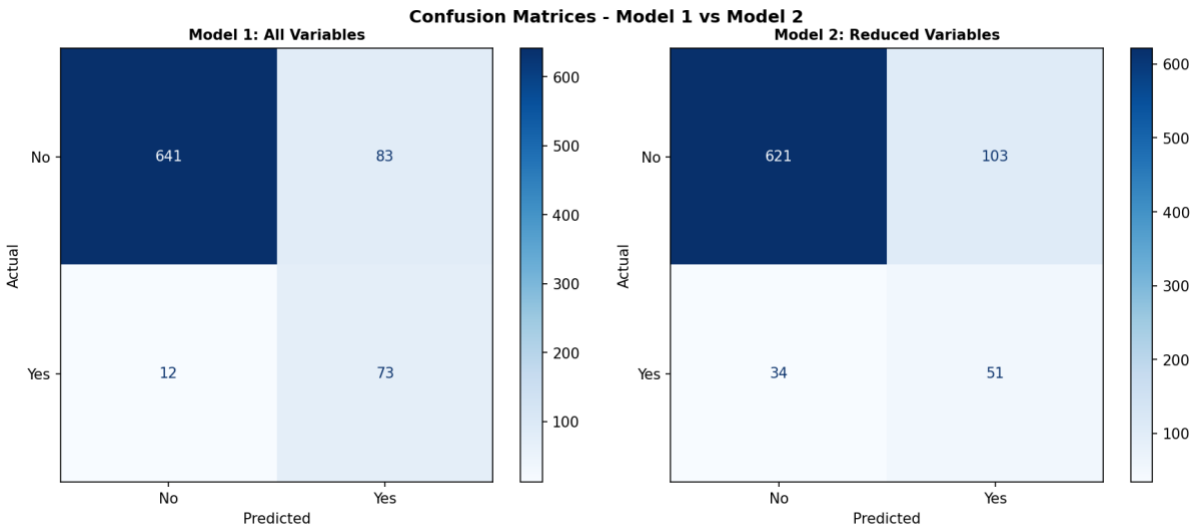

Figure 13 - Confusion Matrices of Model 1 and Model 2

ROC curves also visualise the performance of both models (figure 14). The drop in AUC from ‘model1’ (0.932) to ‘model2’ (0.752) as well was the ROC curve clearly confirms that the variables dropped carried significant predictive power and signifies a lower True Positive Rate (TPR) for ‘model2’.

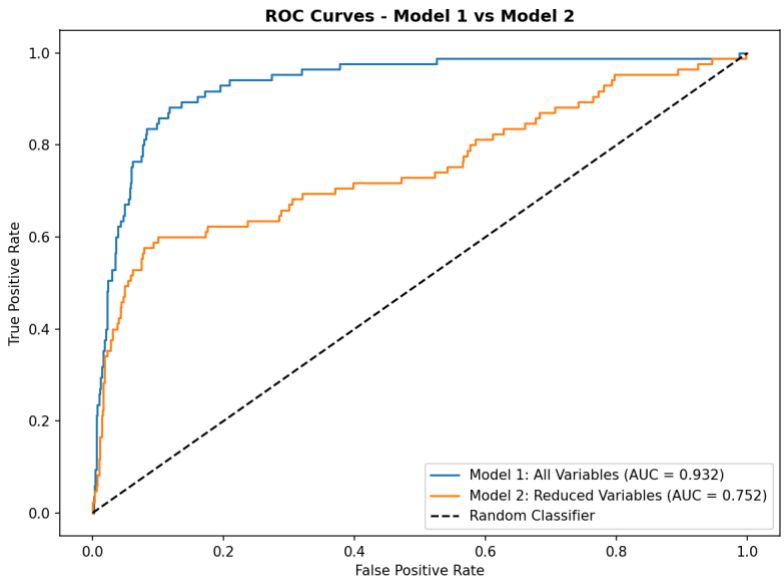

Figure 14 - AUC-ROC Curves of Model 1 and Model 2

Conclusion

Upon evaluation of F1-scores, confusion matrices, ROC curves and AUC values for both predictive ‘model1’ performed better. However, the strong likelihood of leakage from variables such as ‘duration’ means that the model’s scores are inflated. This is a considerable limitation of the initial model.

Additionally, the data was collected during the 2008 to 2013 period, during the global financial crisis and its aftermath. The economic climate shifted across the dataset, so the model may learn patterns from 2009 to predict the outcome of contacts in 2012.

An additional regression model which solely omits ‘duration’ may result in a model which performs better whilst eliminating leakage. Also, as the dataset is complex, a tree-based classifier, such as a Random Forest, may consider non-linear relationships between feature variables and the outcome, resulting in better performance.

Irrespective of the model performance, actionable insights can be drawn from the EDA completed. Prioritising cellular contacts as well as placing calls in March and October increases the likelihood of a client agreeing to open a new deposit account, which can save time and resources for the bank.

References

Bouke, M.A., Zaid, S.A. and Abdullah, A. (2024) 'Implications of data leakage in machine learning preprocessing: a multi-domain investigation'.

Gough, P. and Zhao, J. (2023) 'Data Visualisation', in S. Rowland and L. Kuchel (eds) *Teaching Science Students to Communicate: A Practical Guide*. Cham: Springer International Publishing, pp. 57–68. Available at: https://doi.org/10.1007/978-3-030-91628-2_7.

Moro, S., Cortez, P. and Rita, P. (2014) 'Bank Marketing'. UCI Machine Learning Repository. Available at: https://doi.org/10.24432/C5K306.

PricewaterhouseCoopers (no date) *Data and FinTech in the banking sector: the foundation for innovation*, *PwC*. Available at: https://www.pwc.co.uk/industries/financial-services/fintech/data-and-fintech.html (Accessed: 21 March 2026).

Staal, K. (2023) 'Household Savings and Negative Interest Rates', *International Advances in Economic Research*, 29(1), pp. 1–13. Available at: https://doi.org/10.1007/s11294-023-09870-1.

What is Predictive Modeling? Types & Techniques (no date) *Qlik*. Available at: https://www.qlik.com/us/predictive-analytics/predictive-modeling (Accessed: 21 March 2026).

Appendix

Appendix A: list and description of all dataset variables

- age: (numeric)
- job: type of job (categorical: ’admin.’, ’blue-collar’, ’entrepreneur’, ’housemaid’, ’management’, ’retired’,’self-employed’,’services’,’student’,’technician’,’unemployed’,’unknown’)
- marital: marital status (categorical: ’divorced’, ’married’, ’single’, ’unknown’; note: ’divorced’ means divorced or widowed)
- education: (categorical: ’basic.4y’,’basic.6y’,’basic.9y’,’high.school’, ’illiterate’,’professional.course’,’university.degree’,’unknown’)
- default: has credit in default? (categorical: ’no’,’ yes’, ’unknown’)
- housing: has housing loan? (categorical: ’no’,’ yes’, ’unknown’)
- loan: has personal loan? (categorical: ’no’,’ yes’, ’unknown’)
- Marketing campaign information:
- contact: contact communication type (categorical: ’cellular’, ’telephone’)
- month: last contact month of year (categorical: ’jan’, ’feb’, ’mar’, ..., ’nov’, ’dec’)
- day of week: last contact day of the week (categorical: ’mon’, ’tue’, ’wed’, ’thu’, ’fri’)
- campaign: number of contacts performed during this campaign and for this client (numeric, includes last contact)
- pdays: number of days that passed by after the client was last contacted from a previous campaign (numeric; 999 means client was not previously contacted)

- previous: number of contacts performed before this campaign and for this client (numeric)
- poutcome: outcome of the previous marketing campaign (categorical: ’failure’, ’nonexistent’, ’success’) Social and economic context attributes
- emp.var.rate: employment variation rate - quarterly indicator (numeric)
- cons.price.idx: consumer price index - monthly indicator (numeric)
- cons.conf.idx: consumer confidence index - monthly indicator (numeric)
- euribor3m: euribor 3-month rate - daily indicator (numeric)
- nr.employed: number of employees - quarterly indicator (numeric)
- y: outcome – whether client has agreed to open a long-term deposit account (binary; ‘yes’, ‘no’)